OpenClaw Multimodal Vision

1. Course Content

Course Overview

- OpenClaw's multimodal vision capability provides visual perception through MCP interfaces, enabling robots to "understand" their surroundings
- Covers 3 core vision tools: SeeWhat (capture image), GetBbox (object detection bounding box), GetPlacePoint (get placement point coordinates)
- Supports configuring SeeWhat auto-send channels to automatically push robot observation images to communication channels like Feishu and WhatsApp

Learning Objectives

1. Understand the functionality and use cases of the 3 vision control interfaces provided by OpenClaw
2. Master the configuration method for SeeWhat auto-send channels
3. Master how to invoke vision tools directly via CLI

2. Preparation

- Start the MCP service:

```
ros2 launch dofbot_pro_bringup mcp_service.launch.py
```

- If you need voice broadcast replies, add the `use_tts` parameter: `ros2 launch dofbot_pro_bringup mcp_service.launch.py use_tts:=true`, otherwise it can be omitted

3. Vision-Related Control Interfaces

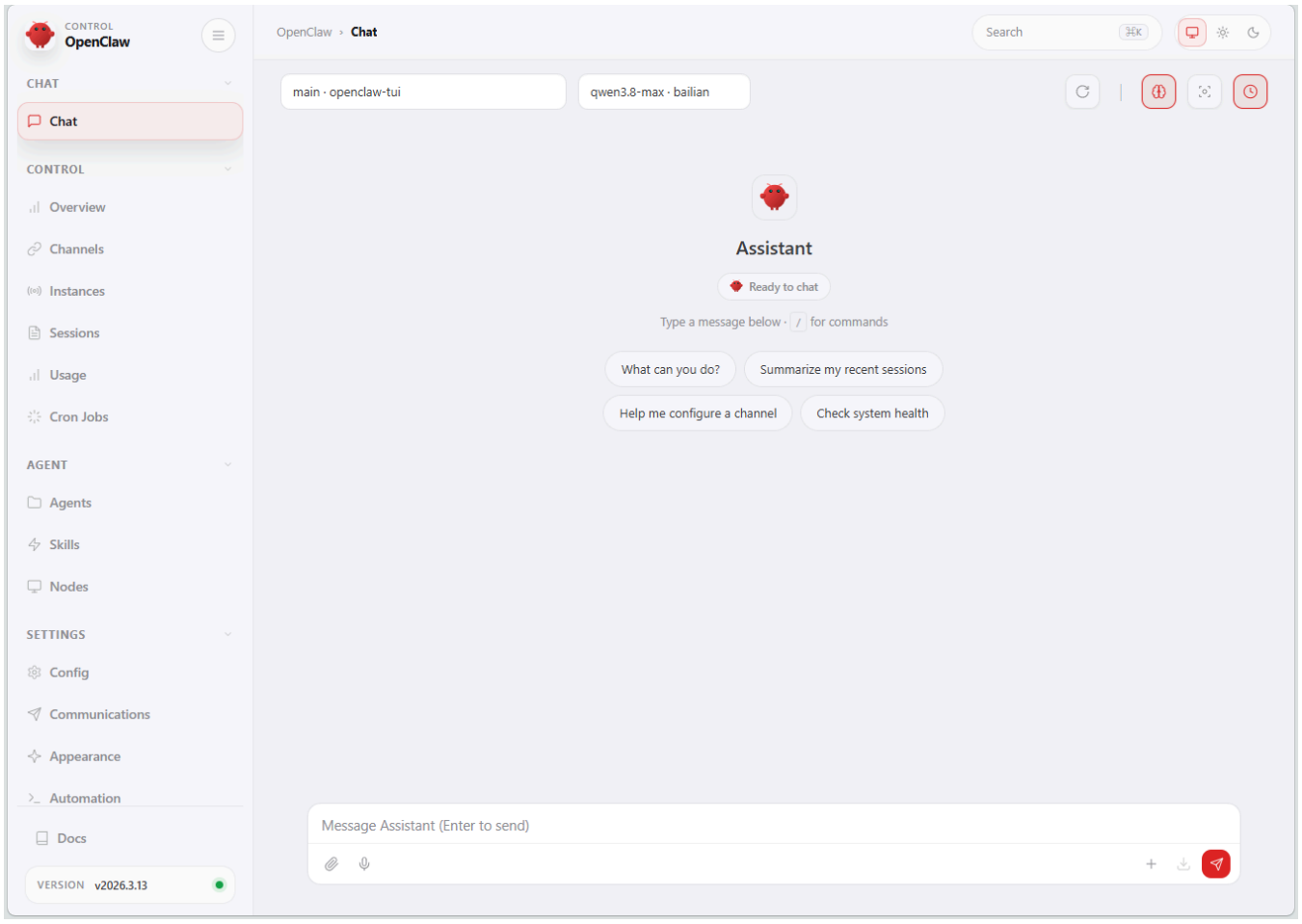
> **Note**
>
> - For detailed functionality, parameters, and usage of the interfaces, please refer to tutorials 【01-OpenClaw Connecting to Robot MCP Service】、 【02-Robot CLI Command Tool】

Tool Name	Function Description
SeeWhat	Capture camera image and return save path
GetBbox	Detect bounding box coordinates of target object
GetPlacePoint	Get placement point coordinates and check reachability

4. Case Demonstration

4.1 Start OpenClaw Interaction

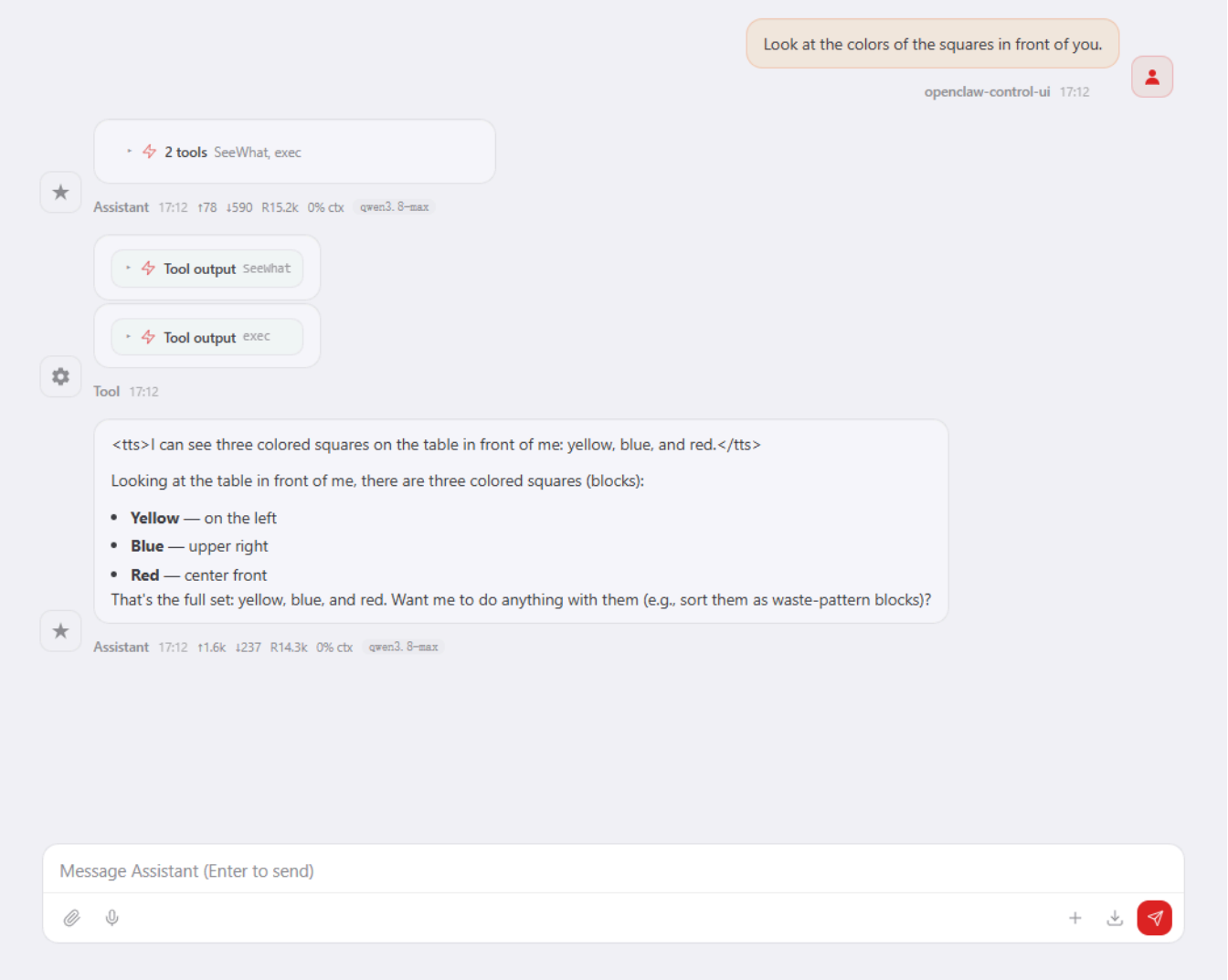
- Use any interaction method to converse with OpenClaw, here demonstrating with Web Chat



> **Tip**
>
> - You can write conversation statements according to your own needs, following is the case demonstration

4.2 Visual Observation and Understanding

- Look at the colors of the squares in front of you.



- When the SeeWhat tool on the MCP server side is called, the image is automatically saved for checking observation content, image path: `$HOME/MODELS/multi_brains_file/image.png`

4.3 Configure SeeWhat Auto-Send Channel

- This feature is used for the robot to automatically send observation perspective images to mobile communication channels when calling SeeWhat to observe the environment
- Configuration file path: `$HOME/.openclaw/openclaw.json`, you can directly modify the configuration file or configure via CLI command, here is an example of CLI command configuration

> **Important**
>
> - SeeWhat_channels is the channel that needs to be auto-sent, images will only be sent to channels in the list

```
...
"robot-control": {
    "enabled": true,
    "config": {
      "SeeWhat_channels": [
        "whatsapp",
        "feishu",
        "weixin"
```

```
      ],
      "feishu_chat_id": "oc_c10ce0aa182cb3f6162d003cxxxxxxx",
      "whatsapp_number": "+8615338xxxxxxx"
    }
  }
...
```

Configuration Parameter Description:

- `SeeWhat_channels`: Auto-send channel list, supports `whatsapp`, `feishu`, `weixin`, when SeeWhat is called, observation images will be automatically sent to channels in the list
- `feishu_chat_id`: Feishu group chat ID, used to specify the send target for Feishu channel
- `whatsapp_number`: WhatsApp recipient phone number, must include country code (e.g., `+86`)

4.3.1 Configure Feishu Channel (China)

```
openclaw config set plugins.entries.robot-control.config.SeeWhat_channels '["feishu"]'
```

- Copy and add Feishu ID





- Set SeeWhat_channels chat ID

```
openclaw config set plugins.entries.robot-control.config.feishu_target
"oc_c10ce0aa182cb3f6162d003c296xxxxx"
```

```
Config overwrite: /home/jetson/.openclaw/openclaw.json (sha256 ad23b53f135f5c020b9c7744e432fc6c5d1b01b2e253dd3b727f47fb5d0ea65d -> 046
7b0c2ed0db483641654f863c5426baa65941449884fff3c1a7c6779ae024e, backup=/home/jetson/.openclaw/openclaw.json.bak)
Updated plugins.entries.robot-control.config.feishu_chat_id. Restart the gateway to apply.
```

- Restart gateway to take effect

```
openclaw gateway restart
# For Raspberry Pi version:
ocg restart
```

- During restart, openclaw logs will indicate the auto-send channel

```
14:27:31 [plugins] robot-control: SeeWhat channels loaded: [feishu]
```

4.3.2 Configure WhatsApp Channel (International)

```
openclaw config set plugins.entries.robot-control.config.SeeWhat_channels
'["feishu","whatsapp"]'
```

- Set SeeWhat_channels phone number

```
openclaw config set plugins.entries.robot-control.config.whatsapp_number "+861533xxxxx"
```

- Restart gateway to take effect

```
openclaw gateway restart
# For Raspberry Pi version:
ocg restart
```

- During restart, openclaw logs will indicate the auto-send channel

```
15:17:28 [plugins] robot-control: SeeWhat channels loaded: [feishu, weixin, whatsapp]
```

> **Note**
>
> - Now you can converse with OpenClaw, and images will be automatically sent to configured communication channels when vision functions are called

4.4 Get Target Bounding Box Coordinates

- Get target object outer bounding box, this tool is used for subsequent robotic arm grasping of objects and tracking targets, GetBbox tool provides the function to obtain target bounding box coordinates based on natural language descriptions
- You can directly call GetBbox via robot_control CLI

```
robot_control call-tool GetBbox --query "yellow block on the table"
```

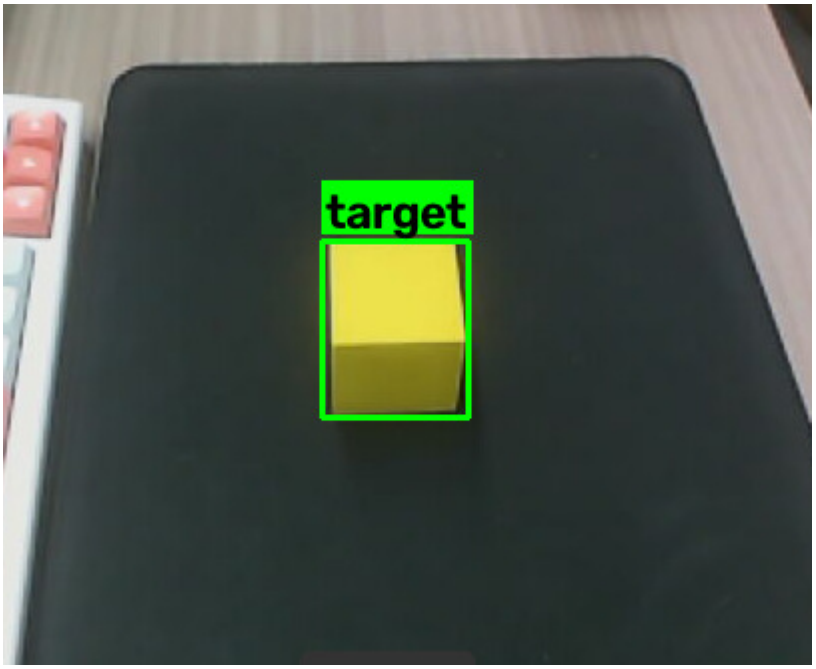
- If the target object exists within the camera's field of view, it will return the bounding box coordinates and the path to the annotated verification image, the bounding box coordinates are used by OpenClaw to decide the next operation.

```
jetson@yahboom:~$ robot_control call-tool GetBbox --query yellowsquareonthetable
{
  "result": {
    "bbox": "{'x1': 392, 'y1': 190, 'x2': 464, 'y2': 269}",
    "verify_image": "/home/jetson/MODELS/multi_brains_file/verify_image.png"
  }
}
```

- Auto-saved verification image path:

```
~/MODELS/multi_brains_file/verify_image.png
```

- You can check through the image whether the vision model accurately draws a bounding box around the target



> **Important**
>
> - The accuracy of visual positioning and bounding box selection depends on the capability of the vision model, which will directly affect robotic arm grasping and visual tracking functions

4.5 Get Placement Point Coordinates (Depth Camera Only)

- This tool is used for **subsequent robotic arm placing objects to specified points**, obtains the spatial 3D coordinates of the target point based on natural language descriptions, and determines whether the point is within the robotic arm's working range
- You can directly call GetPlacePoint via robot_control CLI

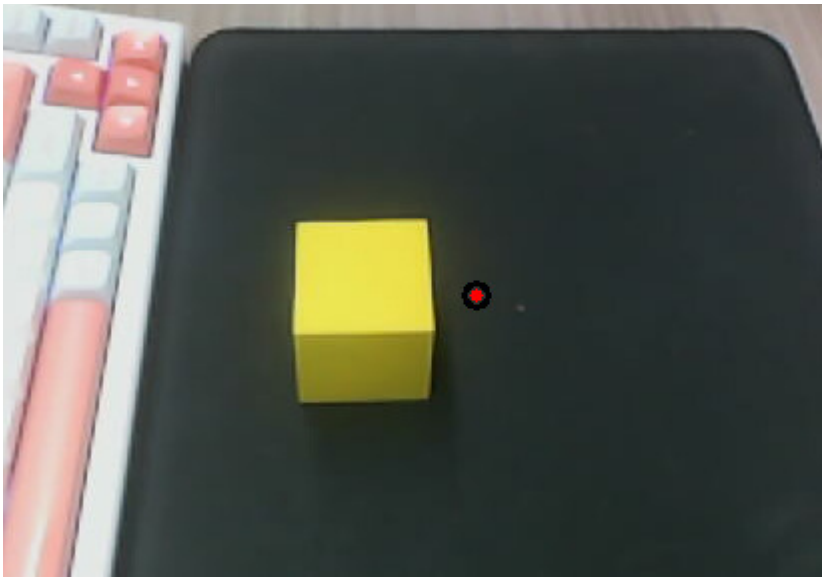
```
robot_control call-tool GetPlacePoint --query "place to the right of the yellow block on the table"
```

- If a point matching the description exists within the camera's field of view, the tool returns the spatial coordinates of the target point in the vehicle's base_link coordinate system, the verification image path, and a notification about whether the point is within the robotic arm's working range

```
jetson@yahboom:~$ robot_control call-tool GetPlacePoint --query placetotherightoftheyellowblockonthetable
{
  "result": {
    "target_point": "{'x': 0.32, 'y': -0.09, 'z': -0.01}",
    "info": "The target point is within the working range of the robotic arm",
    "place_verify_image": "/home/jetson/MODELS/multi_brains_file/place_verify_image.png"
  }
}
```

- Verification image save path:

```
~/MODELS/multi_brains_file/image_point_selected.png
```



5. Source Code Analysis

- Core source code modules:
  - Controller adapter layer: `~/yahboomcar_ros2_ws/dofbot_pro_ws/src/ackermann_mcp/ackermann_mcp/action_controller.py`
  - Vision tool wrapper: `~/yahboomcar_ros2_ws/dofbot_pro_ws/src/ackermann_mcp/ackermann_mcp/vision/visual_tools.py`
  - Object bounding box detection: `~/yahboomcar_ros2_ws/dofbot_pro_ws/src/ackermann_mcp/ackermann_mcp/vision/visual_bounding.py`
  - Grid placement positioning: `~/yahboomcar_ros2_ws/dofbot_pro_ws/src/ackermann_mcp/ackermann_mcp/vision/visual_positioning.py`
  - LLM communication layer: `~/yahboomcar_ros2_ws/dofbot_pro_ws/src/ackermann_mcp/ackermann_mcp/vision/_vision_model.py`

The following is the core implementation and source code analysis of the multimodal vision pipeline.

5.1 `SeeWhat()` — Capture Current Camera Frame

Code Description:

- `SeeWhat`: Unified image capture entry with multi-camera backend adaptive capability
- **Dual backend adaptation**:
  - When working in USB monocular camera mode, captures the latest valid frame from USB video stream and caches to temp directory via `UsbVisionBackend.capture_image()`
  - When working in depth camera mode, calls ROS 2 Trigger client `/save_image` service via `visual_see_what(self)` to trigger camera driver to save and return image physical path

- **Cache maintenance**: Updates `self.image_path` after successful capture, used as input data source by subsequent tools like `GetBbox`, `GetPlacePoint`, `Pick`

```
    def SeeWhat(self) -> list:
        """Capture and save current camera image for vision tools."""
        if self.vision_backend is not None:
            return self.vision_backend.capture_image()
        return visual_see_what(self)
```

Depth camera implementation in `vision/visual_tools.py`:

```
def see_what(node) -> list:
    """Capture and cache the current depth-camera image."""
    response: Trigger.Response = node.image_client.call(Trigger.Request())
    if response.success:
        node.image_path = response.message
        return [True, response.message]

    node.image_path = None
    node.get_logger().error(response.message)
    return [False, response.message]
```

5.2 `GetBbox()` — Get Target Bounding Box Coordinates

Code Description:

- `GetBbox`: Locates target object in scene based on natural language description, returns pixel-level bounding box coordinates `{"x1": ..., "y1": ..., "x2": ..., "y2": ...}`
- **Adaptive pre-refresh**: Automatically calls `SeeWhat()` to refresh frame, ensuring recognition is always based on robot's latest observation
- **Prompt and coordinate system**: Vision LLM uses `ONLINE_PROMPT`, mapping image resolution to standardized scale of `0 ~ 1000`, model outputs normalized rectangular box which is then back-calculated to real pixel coordinate system, greatly improving cross-resolution positioning accuracy and generalization
- **Verification image save**: Draws bounding box around recognition result on original image and saves to `verify_image.png` (USB mode is `verify_image-usb.png`), convenient for review and frontend interface visualization
- **Coordinate safety validation**: Filters noise boxes smaller than threshold via `normalize_image_bbox`, ensuring output bounding box is within current image dimension legal range

```
def get_bbox(node, query: str, capture_image=None) -> list:
    """Capture an image and locate one described object directly."""
    target = str(query).strip()
    if not target:
        return [False, "GetBbox query must not be empty", ""]
    capture_image = capture_image or (lambda: see_what(node))
    capture_result = capture_image()
```

```
    if not capture_result[0]:
        return [False, capture_result[1], ""]

    source_image = cv2.imread(str(node.image_path))
    if source_image is None:
        return [False, f"Failed to read camera image: {node.image_path}", ""]
    verify_path = _artifact_path(node, "verify_image_path", "verify_image.png")
    try:
        tool = node._get_bbox_tool()
        with node._visual_tool_lock:
            result = tool.BoundingBoxDetection(
                image_path=str(node.image_path),
                target_description=target,
                output_path=verify_path,
            )
    except Exception as exc:
        node.get_logger().error(f"GetBbox failed: {exc}")
        return [False, f"GetBbox failed: {exc}", ""]

    if result.error:
        node.get_logger().error(result.error_message)
        return [False, f"GetBbox failed: {result.error_message}", ""]
    if not result.found:
        return [False, f"GetBbox: target '{target}' not found in the image", ""]

    height, width = source_image.shape[:2]
    bbox = normalize_image_bbox(
        {
            "x1": result.x1,
            "y1": result.y1,
            "x2": result.x2,
            "y2": result.y2,
        },
        image_width=width,
        image_height=height,
        min_size=2,
    )
    if bbox is None:
        return [False, "GetBbox returned an invalid image bounding box", ""]
    x1, y1, x2, y2 = bbox
    node.get_logger().info(f"GetBbox: target bbox=({x1},{y1},{x2},{y2})")
    return [True, str({"x1": x1, "y1": y1, "x2": x2, "y2": y2})]
```

5.3 `GetPlacePoint()` — Get Placement Point 3D Space Coordinates

Code Description:

- `GetPlacePoint`: For depth camera scenarios, calculates 3D space coordinates `(x, y, z)` of free placement point based on natural language (e.g., "place on the empty space to the right of the yellow block"), and checks if it is within the robotic arm's working envelope
- **Grid overlay positioning mechanism**: Overlays semi-transparent grid with row/column numbers and

top/bottom/left/right orientation text labels on original image with spacing of `place_point_grid_size` (default 20 pixels) via `draw_grid_overlay`, vision model directly outputs candidate grid cell `{"found": true, "col": <int>, "row": <int>}`, greatly reducing hallucination rate of vision LLM directly predicting absolute pixel coordinates

- **3D space projection resolution**: After extracting grid center pixel point, calls depth camera's `/get_target_pose` service via ROS 2, uses pixel sampling radius `target_circle` combined with depth point cloud to resolve actual 3D position in robot's `base_link` coordinate system
- **Working range determination**: Compares robotic arm limit working range `place_distance` (X-axis/Y-axis allowed boundaries), returns out-of-range warning if exceeded, returns placement confirmation if within range

```
def get_place_point(node, query: str) -> list:
    """Locate a free placement pixel, then convert it to a 3D target pose."""
    target = str(query).strip()
    if not target:
        return [False, "GetPlacePoint query must not be empty", ""]
    if not node._prepare_chassis_for_place_point():
        return [False, "Failed to adjust chassis before locating place point", ""]

    capture_result = see_what(node)
    if not capture_result[0]:
        return [False, f"SeeWhat error:{capture_result[1]}", ""]
    source_image = cv2.imread(str(node.image_path))
    if source_image is None:
        return [False, f"Failed to read camera image: {node.image_path}", ""]

    verify_path = _artifact_path(node, "place_verify_image_path",
"place_point_selected.png")
    try:
        tool = node._get_place_point_tool()
        with node._visual_tool_lock:
            result = tool.FindTargetPoint(
                image_path=str(node.image_path),
                target_description=target,
                output_path=verify_path,
            )
    except Exception as exc:
        node.get_logger().error(f"GetPlacePoint failed: {exc}")
        return [False, f"GetPlacePoint failed: {exc}", ""]

    if result.error:
        node.get_logger().error(result.error_message)
        return [False, f"GetPlacePoint failed: {result.error_message}", ""]
    if not result.found:
        return [False, "Point is empty; the target point may not exist in the frame.", ""]

    height, width = source_image.shape[:2]
    pixel = normalize_image_point({"x": result.x, "y": result.y}, image_width=width,
image_height=height)
    if pixel is None:
        return [False, "GetPlacePoint returned an invalid image point", ""]
```

```
    x, y = pixel

    if not node.get_pose_client.wait_for_service(timeout_sec=1.0):
        return [False, "GetTargetPose service is unavailable", ""]
    request = GetTargetPose.Request(x=float(x), y=float(y), radius=node.target_circle)
    try:
        pose_result: GetTargetPose.Response = node.get_pose_client.call(request)
    except Exception as exc:
        return [False, f"GetTargetPose call failed: {exc}", ""]
    if not pose_result.success:
        return [False, str(pose_result.message), ""]

    position = pose_result.pose.position
    point = str({"x": round(position.x, 2), "y": round(position.y, 2), "z":
round(position.z, 2)})
    node.get_logger().info(f"Target Point: x:{position.x:.2f}, y:{position.y:.2f}, z:
{position.z:.2f}")

    if abs(position.x) > node.place_distance[2] or abs(position.y) >
node.place_distance[3]:
        return [False, "The target point is outside the working range of the robotic arm",
point]
    return [True, "The target point is within the working range of the robotic arm", point]
```

5.4 Vision Bottom-Layer Model Communication and Prompt Architecture

Source code module paths:

- `~/yahboomcar_ros2_ws/dofbot_pro_ws/src/ackermann_mcp/ackermann_mcp/vision/visual_bounding.py`
- `~/yahboomcar_ros2_ws/dofbot_pro_ws/src/ackermann_mcp/ackermann_mcp/vision/visual_positioning.py`
- `~/yahboomcar_ros2_ws/dofbot_pro_ws/src/ackermann_mcp/ackermann_mcp/vision/_vision_model.py`

5.4.1 Standard OpenAI Compatible Adapter (`_vision_model.py`)

The bottom layer eliminates third-party complex dependencies and directly wraps a unified client call and JSON parsing mechanism based on the official standard `openai` Python SDK:

```
from openai import OpenAI

def complete(*, model: str, api_base: str, api_key: str, messages: list[dict[str, Any]],
timeout_sec: float):
    """Call configured standard OpenAI-compatible multimodal LLM endpoint"""
    client = OpenAI(
        api_key=api_key,
        base_url=api_base,
        timeout=float(timeout_sec),
```

```
        max_retries=1,
    )
    return client.chat.completions.create(
        model=normalize_model_name(model),
        messages=messages,
    )

def parse_json_response(response: Any) -> dict[str, Any]:
    """Parse JSON object returned by model, automatically remove Markdown code block
wrapping"""
    content = response.choices[0].message.content
    text = content.strip()
    if text.startswith("```"):
        lines = text.splitlines()
        if lines and lines[0].strip().lower() in {"```", "```json"}:
            lines = lines[1:]
        if lines and lines[-1].strip() == "```":
            lines = lines[:-1]
        text = "\n".join(lines).strip()
    return json.loads(text)
```

5.4.2 Object Bounding Box Detection and Prompt Design (`visual_bounding.py`)

- **Normalized prompt (`ONLINE_PROMPT`)**:
  Maps images of any resolution to a standardized coordinate system of `[0, 1000]`, strictly constraining model output to `{"found": true, "x1": <int>, "y1": <int>, "x2": <int>, "y2": <int>}`.
- **Online detection execution (`_online_detection`)**:

```
    def _online_detection(self, image_path: str, target_description: str) -> VisualResult:
        data_url = encode_image_data_url(image_path)
        system_prompt = Template(ONLINE_PROMPT).substitute(width=1000, height=1000)
        try:
            response = complete(
                model=self.model,
                api_base=self.base_url,
                api_key=self.api_key,
                messages=[
                    {"role": "system", "content": system_prompt},
                    {
                        "role": "user",
                        "content": [
                            {"type": "text", "text": f"Target: {target_description}"},
                            {"type": "image_url", "image_url": {"url": data_url}},
                        ],
                    },
                ],
                timeout_sec=self.timeout_sec,
            )
            data = parse_json_response(response)
        except Exception as exc:
            return VisualResult(error=True, error_message=f"Vision model error: {exc}")
```

```
        if not data.get("found"):
            return VisualResult(found=False, error_message=f"Target '{target_description}'
not found")

        # Convert [0, 1000] normalized coordinates back to original image pixel coordinates
        return VisualResult(
            found=True,
            x1=int(round(data["x1"] / 1000.0 * img_w)),
            y1=int(round(data["y1"] / 1000.0 * img_h)),
            x2=int(round(data["x2"] / 1000.0 * img_w)),
            y2=int(round(data["y2"] / 1000.0 * img_h)),
        )
```